

MAX NICOSIA, PhD

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PROFESSIONAL PROFILE

Software R&D and Systems Engineer with over 10 years of experience designing, prototyping, implementing and validating complex technical systems across simulation, XR, safety-critical software, distributed systems, hardware/software integration and low-latency applications.

Experienced in taking early-stage technical concepts from requirements and architecture through implementation, experimentation, optimisation and delivery. Background includes technical leadership at Woven by Toyota, production low-latency C++ systems, UK Ministry of Defence-funded R&D, XR and simulation platforms, sensor-integrated middleware, and human-in-the-loop systems.

Strong hands-on development background in C++, C# and Python, with additional experience across TypeScript/JavaScript, distributed and networked systems, real-time visualisation, automated testing and research validation.

CORE EXPERTISE

R&D & Rapid Prototyping • Systems Architecture & Design • Simulation • XR / AR / VR • Safety-Critical Systems • Human-in-the-Loop Systems • Verification & Validation • Hardware/Software Integration • Distributed Systems & Middleware • Real-Time & Low-Latency Software • Performance Optimisation • Technical Leadership

TECHNICAL SKILLS

Languages: C / C++ • C# • Python • Java • TypeScript • JavaScript • Ruby • R • SQL

Systems & Software Engineering: Multithreading • Distributed Systems • Middleware • Networking • TCP/IP • WebSockets • Windows/Linux Sockets • CAN • APIs • Dependency Injection • Automated Testing • CI/CD • Performance Profiling • Low-Latency / Real-Time Software

Simulation / XR: Unity • Unreal Engine • AR / VR / XR • Vehicle & Physics Simulation • 3D Visualisation • Human-in-the-Loop Systems • Eye Tracking • Sensor Integration • AI Pathfinding • Shaders / Filters • Mixamo

Parallel / High-Performance Computing: OpenMP • MPI • pthreads • Lockless Programming • Memory Barriers • Ring Buffers • Memory Pools • Cache Alignment • Compiler Optimisation

Data / Scientific / ML: NumPy • SciPy • Pandas • scikit-learn • PyTorch • Jupyter • d3.js • Bayesian Optimisation • Data Analysis • Experimental Modelling

Tools / Platforms: Docker • Podman • CMake • Make • GitHub Actions • Bash • PowerShell • Gradle • AWS • Android • iOS • Xcode

PROFESSIONAL EXPERIENCE

Woven by Toyota — Tech Lead, XR Safety / Driver Sync Assists (Jun 24 – Mar 26)

Tokyo, Japan

- Technical lead for the judgement and simulation workstream, directly managing up to 2 engineers and coordinating priorities, task allocation, code reviews, mentoring and technical direction.
- Re-architected an inherited collection of Unity scenes and assets into a structured C# simulation platform suitable for repeatable experimentation, safety validation and product development.
- Restructured the codebase around interfaces, inheritance, dependency injection and the Humble Object pattern, improving maintainability and enabling automated testing of previously difficult-to-test simulation and judgement logic.
- Developed automated testing covering shared physics calculations and safety judgement for vehicles, pedestrians, traffic lights and other obstacles.
- Profiled and optimised the Unity simulation, eliminating repeated execution and other bottlenecks to achieve the 60+ FPS target, typically reaching ~90–120 FPS depending on scene complexity while driving 5 × 4K displays plus 2 × HD mirrored displays.
- Designed and implemented driver-safety judgement using live and simulated data, including probabilistic collision risk, driver reaction time and cognitive-load considerations across vehicles, pedestrians, traffic lights, road layouts and other hazards.
- Developed reproducible vehicle and pedestrian AI/pathfinding and integrated Vehicle Physics Pro for realistic vehicle and engine simulation.
- Developed capture, playback and re-simulation capabilities to support repeatable testing, experimentation and analysis.
- Developed C/C++ wrappers and CAN interfaces for steering, speed, acceleration, pedals and other vehicle-control inputs.
- Supported 16-participant experiments using Latin-square counterbalancing for verification and validation of judgement and safety logic.
- Defined testing and product-verification strategies and produced documentation supporting PoC and beta-gate approval.

- Presented quarterly technical demonstrations and project progress to product management.
- Developed visualisation technology resulting in patent filings in Japan and the US; named as co-inventor.
- Delivered project milestones through PoC verification and beta gates, progressing the project into early production development.

FinstadiumX — Strategic Development Team Lead (Apr 24 – May 24)

Tokyo, Japan

- Led and mentored a team of 2 engineers developing and maintaining production low-latency C++ financial transaction systems.
- Designed, implemented and deployed C++ components for high-frequency equities, futures and dark-pool transaction processing on Linux.
- Worked on production systems handling approximately 2,000–5,000 transactions per second at ~20 ms latency and transaction flows associated with approximately ¥7–10 billion in daily revenue per product.
- Applied low-latency techniques including memory barriers, lockless data structures, ring buffers, memory pools, cache alignment and compiler optimisation.
- Profiled legacy components and began formalising latency and performance testing for production systems.
- Developed build procedures across Linux and Windows using CMake and Make and maintained Docker/Podman development, testing and production environments.
- Maintained FIX compliance and technical documentation.
- Introduced automated CMake build, test and deployment scripts that prevented defective changes from reaching production.

Tokyo Academics — Head of Research (Nov 22 – Mar 24)

Tokyo, Japan

- Managed approximately 20 part-time researchers supervising student research projects and oversaw research operations, sales and marketing strategy.
- Supervised computer-science projects involving software prototyping and machine learning across multiple technologies.
- Grew the research programme from approximately 20 to 50–60 students year-on-year.
- Achieved approximately 125% year-on-year growth in sales receipts against a 50% growth target.

Cambridge Intelligent Systems UK Ltd — CEO, Co-founder & Technical Lead (Nov 17 – Jun 25)

Cambridge, UK

R&D consultancy delivering UK Ministry of Defence research programmes in XR, simulation, human factors and safety-critical systems.

- Co-founded and technically led a two-person R&D company, securing approximately £150,000 in combined funding across two UK MoD programmes.
- Served as principal engineer and sole software developer, taking projects from proposal and stakeholder requirements through architecture, implementation, integration, experimentation, demonstration and final delivery.
- Worked continuously with MoD/DSTL stakeholders throughout project inception, requirements definition, development and evaluation.
- Managed milestones, budgets, procurement, proposals and company operations.
- Personally presented and demonstrated completed systems to senior military stakeholders.

Project ACC2006330 — Attention-Aware System for Mixed and Augmented Reality Controls in Armoured Vehicles (Mar 20 – Feb 21)

- Designed and developed a complete Unity/C# XR simulation for testing and training operator threat detection using attention-aware visualisations.
- Built configurable training stages, NPCs, animations and navigation/pathfinding, NATO-compliant threat visualisations, 3D/minimap displays, 360-degree views, threat-tagging interfaces and performance logging.
- Implemented dynamic attention-aware techniques including attended/unattended region tracking, animated guidance and out-of-view threat indicators.
- Personally delivered demonstrations and presentations to MoD stakeholders.
- Software was subsequently incorporated into tank-operation training practices.

Project ACC2000981 — Mixed Reality Controls for Armoured Vehicles (Dec 18 – Oct 19)

- Designed and developed a complete Unity/C# VR prototype emulating an augmented-reality control environment inside an armoured vehicle.
- Developed GVA-compliant interfaces, 360-degree simulated camera views, UltraLeap hand interaction and NATO-compliant symbology.
- Investigated replacement of physical controls with mixed-reality interaction to reduce hardware requirements and vehicle weight.
- Personally delivered demonstrations to senior military stakeholders; concepts subsequently informed future vehicle-development work.

Cambridge Technical Services Ltd — Consultant (May 17 – Dec 17)

Cambridge, UK

Project ACC101965 — Distributed Multi-Display Middleware for Operator Gaze and Attention Tracking

- Adapted the distributed attention-management architecture developed during doctoral research into a separate safety-critical demonstrator.
- Deployed the system across 5 application computers and 5 eye trackers with central coordination and time-synchronisation services.
- Developed C++ middleware for processing and distributing operator-attention data across networked applications and a purpose-built API for application integration.
- Developed TypeScript/JavaScript/d3.js visualisation for live attention state, history and missed-information monitoring.
- Demonstrated the system as a potential enhancement to RAF air-traffic-control systems.

University of Cambridge, Department of Engineering — Research Associate (Oct 19 – Jul 20)

Cambridge, UK

- Supported research software and experimental infrastructure while continuing doctoral-system development and validation.
- Developed and maintained support software in C++, TypeScript and Bash, alongside database maintenance, scripting and experimental data collection.

EDUCATION & RESEARCH

PhD Engineering, University of Cambridge (2014–23)

Thesis: *Design, Implementation and Evaluation of an Attention Management System*

- Designed, implemented and validated a distributed attention-management architecture for complex multi-display, multi-task human-in-the-loop systems.
- Developed four integrated software systems spanning the sensor system, test application, simulator and attention-management system.
- Built multithreaded C++ sensor drivers and middleware interfacing with Tobii eye trackers, processing gaze/head-tracking data and routing state and messages across networked applications.
- Developed a multithreaded Python management system responsible for global state aggregation, performance baselines and indicators, intervention generation and scheduling.
- Developed APIs and integration mechanisms allowing external applications to interact with the middleware.
- Engineered the architecture around hardware limitations requiring independent eye-tracker/display computers with central coordination and time synchronisation.
- System architecture supported up to 8 application machines, 8 Kinect sensors and 8 eye trackers plus coordination and time-server infrastructure; the final experimental deployment used 4 application/eye-tracker machines plus coordination and time servers.

- Designed and conducted three iterative studies involving participant testing, system verification and simulation-based validation, with cohorts of 16, 8 and 16 participants.
- Demonstrated statistically significant improvements in primary and secondary task performance in 3 of 4 trials using adaptive intervention strategies.
- Research produced multiple peer-reviewed publications and subsequently formed the basis for external MoD/RAF demonstrator work.

MPhil Advanced Computer Science, University of Cambridge (2012–13)

Thesis: *A Privacy-Preserving Advertisement Delivery System*

- Developed a Python-based advertisement-delivery server and Chrome extension using publish-subscribe communication, WebSockets and RSA-based privacy mechanisms.
- Coursework included Security, Data-Centric Networks, Network Theory, Distributed Networks and Mobile Application Development.

BSc Computer Science, University of St Andrews (2008–12)

- Dissertation involved implementation of a reduced INLIPv4 network protocol in C as a Linux kernel-level module.

EARLIER RESEARCH & TECHNICAL EXPERIENCE

University of St Andrews — Research Assistant (Jan 12 – Sep 12)

Developed C#/XNA experimental software for research into user-performance perception using Microsoft Surface hardware; performed modelling and analysis in R. Research resulted in an ACM CHI 2014 publication.

iGEM — Research Internship (Jun 11 – Sep 11)

Worked on biological experimentation and MATLAB reaction-rate modelling as part of a SULSA/Wellcome Trust-funded synthetic-biology project that received an iGEM Gold Medal.

University of St Andrews — Research Assistant (May 11 – Jun 11)

Developed Python simulation software for modelling mosquito disease transmission and resistance and performed automated data processing/modelling using Python, R, Bash and AWK; also contributed to seal-tracking and clustering research.

Technical University of Denmark — C Developer (Jun 10 – Aug 10)

Developed Linux accessibility functionality for the Evince/GNOME ecosystem, including integration with Orca and accessibility documentation.

University of St Andrews — Software Developer Intern (Aug 10 – Sep 10)

Developed functionality for the OpenMath binary encoding package for GAP; the package was subsequently published in the GAP ecosystem.

PATENTS

User Status Notification Method and User Support System — Co-inventor

Japanese Application No. 2025-063631, filed Apr 25 | Related US application, filed Jan 26

PUBLICATIONS

- Nicosia, M. & Kristensson, P. O. (2024). *Risk management in human-in-the-loop AI-assisted attention aware systems*. Academic Press.
- Nicosia, M. & Kristensson, P. O. (2021). *Design principles for AI-assisted attention aware systems in human-in-the-loop safety critical applications*. Springer Nature.
- Nicosia, M. & Kristensson, P. O. (2020). *A conceptual design of an inattention management middleware with adaptive target saliency*. IEEE Aerospace Conference.
- Nicosia, M. & Kristensson, P. O. (2018). *Inattention management middleware for human-in-the-loop multi-display applications*. IEEE HCCS.
- Nicosia, M., Oulasvirta, A. & Kristensson, P. O. (2014). *Modeling the perception of user performance*. ACM CHI.

ADDITIONAL PROJECT

Developed an iOS mobile application prototype in Unreal Engine using Bayesian optimisation to adapt learning experiences based on user performance.

LANGUAGES

English: Native | **Spanish:** Fluent | **German:** Business | **Japanese:** Basic